1.2.9★ How many sets of four consecutive positive integers are there such that the product of the four integers is less than 100,000? **Hints:** 123, 104

1.3.4★ At the Gooddog Obedience School, dogs can learn to do three tricks: sit, stay, and roll over. Of the dogs at the school:

50 dogs can sit	17 dogs can sit and stay
29 dogs can stay	12 dogs can stay and roll over
34 dogs can roll over	18 dogs can sit and roll over
9 dogs can do all three	9 dogs can do none

(a) How many dogs are in the school? **Hints:** 141

(b) How many dogs can do exactly 2 tricks? **Hints:** 131

1.28 In the upcoming election, voters can choose:

- Abe or Barb for President;
- Carlos, Darla, or Eve for Vice-President;
- Fred, Gina, or Hal for Secretary.

(b) How many different ways can the ballot be completed if each voter must cast a vote for each office?

(c) How many ways can the ballot be completed if the voter may choose not to vote in some (or all) of the elections?

1.29 How many positive integers less than 500 can be written as the sum of two positive perfect cubes? **Hints:** 35, 65

1.32★ In my fencing club there are twice as many boys who are in high school as there are boys who are in middle school. Of the students in the club who are in middle school, there are three times as many girls as there are boys. Half of the girls who are in the club are in high school. If there are 72 people in my fencing club, how many are middle school boys? **Hints:** 49, 4

Problem 2.5: The inhabitants of the island of Mumble (from Problem 2.2) decided to expand their alphabet to the standard Roman alphabet (26 letters, A through Z). But they still limit their words to 3 letters or less, and for some reason, they insist that all words contain the letter A at least once. How many 3-letter words are possible?

Problem 2.6: How many three-digit numbers are not multiples of 7?

Problem 2.7: The Smith family has 4 sons and 3 daughters. In how many ways can they be seated in a row of 7 chairs such that at least 2 boys are next to each other?

Problem 2.10: In how many ways can we pick a group of 3 different numbers from the group

$$1, 2, 3, \ldots, 500$$

such that one number is the average of the other two? (The order in which we pick the numbers does not matter.)

Problem 2.14: Our math club has 20 members and 3 officers: President, Vice President, and Treasurer. However, one member, Ali, hates another member, Brenda. How many ways can we fill the offices if Ali refuses to serve as an officer if Brenda is also an officer?

Problem 2.15: We have the same club (20 members and 3 officers), but this time, Ali has a huge crush on Brenda, and won't be an officer unless she is one too. Brenda is unaware of Ali's affections, and doesn't care if he is an officer or not; she's perfectly happy to be an officer even if Ali isn't one. In how many ways can the club choose its officers?

2.5.3 The Jones family has 5 boys and 3 girls, and 2 of the girls are twins. In how many ways can they be seated in a row of 8 chairs if the twins insist on sitting together, and their other sister refuses to sit next to either of her sisters? **Hints:** 53

2.5.4 Our club has 20 members, 10 boys and 10 girls. In how many ways can we choose a president and a vice-president if:

(a) we have no restrictions other than the same person can't hold both offices?

2.5.5 Our club has 25 members, and wishes to pick a president, secretary, and treasurer. In how many ways can we choose the officers, if:

(c) individual members are allowed to hold 2, but not all 3, offices? **Hints: 161**

2.18 How many different three-digit security codes are possible using the digits 1-5, if the second digit cannot be the same as the first, and third digit cannot be the same as the second? *(Source: MATHCOUNTS)*

2.19 How many triples (a, b, c) of even positive integers satisfy $a^3 + b^2 + c \leq 50$?

2.23 How many 4-digit numbers have the second digit even and the fourth digit at least twice the second digit?

2.26 How many positive, even three-digit numbers exist such that the sum of the hundreds digit and the tens digit equals the units digit? *(Source: MATHCOUNTS)*

2.29 The n members of a committee are numbered 1 through n. One of the members is designated as the "Grand Pooh-Bah." The n members sit in a row of n chairs, but no member with a number greater than the Grand Pooh-Bah may sit in the seat to the immediate right of the Grand Pooh-Bah. Suppose that the Grand Pooh-Bah is member number p, where $1 \leq p \leq n$. Find a formula, in terms of n and p, for the number of ways for the committee to sit. **Hints:** 220 *(Source: MATHCOUNTS)*

2.30$\star$ In how many ways can you spell the word NOON in the grid below? You can start on any letter, then on each step you can move one letter in any direction (up, down, left, right, or diagonal). You cannot visit the same letter twice.

NNNN
NOON
NOON
NNNN

2.31 A palindrome is a number that reads the same forwards and backwards, such as 12321.

(d)$\star$ Suppose we form all k-digit palindromes that consist only of 8's and 9's such that there is at least one of each digit. What is the smallest value of k such that there are at least 2004 numbers in the list? **Hints:** 46

2.32 Let S be a cube. Compute the number of planes which pass through at least three vertices of S. **Hints:** 152, 68

2.33★ How many zeroes do we write when we write all the integers from 1 to 256 in binary? **Hints:** 178 *(Source: ARML)*

Problem 3.8: A *convex* polygon is a polygon in which every interior angle is less than 180 degrees. A *diagonal* of a convex polygon is a line segment which connects two non-adjacent vertices.

(c) How many diagonals does a convex hexagon have?

(d) Find a formula for the number of diagonals of a convex polygon with n sides, where n is any positive integer greater than 2. Make sure that your formula matches your answers from parts (a)–(c) where $n = 4, 5, 6$.

Problem 3.9: In how many different ways can 6 people be seated at a round table? Two seating arrangements are considered the same if, for each person, the person to his or her left is the same in both arrangements. In other words, the two arrangements in Figure 3.3 are the same.

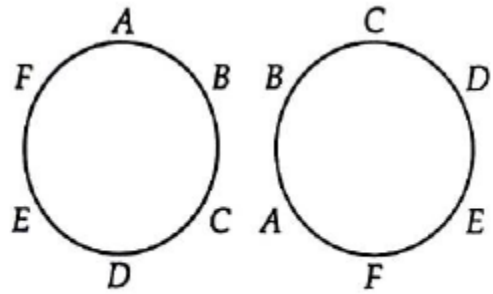

Figure 3.3: Two identical 6-person seatings

Problem 3.10: In how many distinct ways can 4 keys be placed on a keychain? Two arrangements are not considered different if the keys are in the same order (or can be made to be in the same order without taking the keys off the chain).

3.4.3 A Senate committee has 5 Democrats and 5 Republicans. In how many ways can they sit around a circular table:

(c) if each member sits next to two members of the other party? **Hints:** 194

3.4.5★ In how many ways can we seat 8 people around a circular table if Alice and Bob won't sit next to each other? **Hints:** 71

3.18 There are 6 married couples at a party. At the start of the party, every person shakes hands once with every other person except his or her spouse. How many handshakes are there?

3.25 Suppose $j < k$ are positive integers. Find a formula for the sum of all of the consecutive integers from j to k inclusive; that is, find a formula for

$$j + (j+1) + \cdots + (k-1) + k.$$

3.26★ A regular tetrahedron is a triangular pyramid whose faces are all equilateral triangles. How many distinguishable ways can we paint the four faces of a regular tetrahedron with red, blue, green, and orange paint such that no two faces are the same color? **Hints:** 166

3.27★ I have 7 different keys: 2 for my house, 2 for my car, and 3 for my office. I want to place these keys on a keyring such that the house keys are next to each other and the car keys are next to each other. In how many ways can I place the keys on the keyring? (As usual, rotations and flips of the keyring don't change the arrangement.)

4.2.3 Our water polo team has 15 members. I want to choose a starting lineup consisting of 7 players, one of whom will be the goalie (the other six positions are interchangeable). In how many ways can I choose my starting lineup? **Hints:** 88

4.2.4 Consider a regular octagon. How many triangles can be formed whose vertices are the vertices of the octagon? **Hints:** 120

4.3.2 What is $\binom{n}{0}$ for any positive integer n?

4.3.5 Does $\binom{5}{7}$ make any sense? Why or why not? **Hints:** 56

4.14 A lock has 5 buttons numbered 1-5. The lock is opened by pushing two buttons simultaneously and then pushing one button alone. How many combinations are possible?
(Source: MATHCOUNTS)

4.20 The Aviation Club at Rocco's school has 12 members (including Rocco). They need to choose a 3-person Helicopter Committee and a 4-person Glider Committee. Students can serve on either or both committees, but Rocco refuses to serve on both—he will only serve on one or the other (or neither). In how many ways can both committees be chosen?

4.22 There are 5 different pairs of gloves, where left and right are distinguishable. Select 4 from the 10 gloves.

(b) How many ways are there to select 4 such that some 2 of the 4 make a pair?

4.23★ In poker, a 5-card hand is called a *three of a kind* if there are three cards of one rank and two other cards which are not the same rank as each other or as the other three cards. How many 5-card hands are three of a kind? (Assume we are using a standard 52-card deck with 13 ranks in each of 4 suits.)
Hints: 133, 214